Zhao Huang

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EDUCATION BACKGROUND

ETH Zurich **Zurich, Switzerland, 09/2023 – 06/2026**

- Master of Computer Science (Visual and Interactive Computing)
- Grade: 5.4/6 (expected, 6 is highest)

The University of New South Wales **Sydney, Australia, 06/2020 – 01/2023**

- Bachelor of Computer Science with High Distinction
- Grade: 85.2% (GPA: 3.96/4); Ranking: Top 5%
- Honor: Dean's Honours List for 2021 (Highly Commended); Dean's Honours List for 2022 (Highly Commended)

PAPERS

*Equal Contribution

- REACT3D: Recovering Articulations for Interactive Physical 3D Scenes
  Zhao Huang, Boyang Sun, Alexandros Delitzas, Jiaqi Chen, Marc Pollefeys
  – *IEEE Robotics and Automation Letters (RA-L), 2026* [Computer Vision, Robotics]

- CoMind: Understanding Collaborative Human Activity from Multiple Minds and Views
  Alexey Gavryushin*, Dingxi Zhang*, **Zhao Huang***, Alexandros Delitzas, Jiaqi Chen, Ben Ellis, Cedric Zöllner, Manthan Patel, Manuel Kaufmann, Marc Pollefeys, Xi Wang
  – *The 19th European Conference on Computer Vision (ECCV), 2026* [Multi-modal Learning, VLM]

- Automating the Visual Inventory of Building Products Using a Hybrid Computer Vision–Language Model Pipeline
  Shuting Mi, Xiaofei Wu, Shunchang Liu, **Zhao Huang**, Catherine De Wolf, Mennatallah El-Assady, Pei-Yu Wu
  – *Developments in the Built Environment (DIBE), 2026* [Computer Vision, Deep Learning]

PATENTS

- Automatically Determining a Skeleton Line and Airbridges of Coplanar Waveguide
- Method and Apparatus for Air Bridge Arrangement in Circuit Layout, Device, Medium, and Product

RESEARCH EXPERIENCE

Recovering Articulations for Interactive Physical 3D Scenes, Semester Project in ETH CVG **09/2024 – 10/2025**

Supervisor: *Prof. Dr. Marc Pollefeys*, *Boyang Sun*, Alexandros Delitzas, Jiaqi Chen **Role:** *First Contributor*

Project Introduction: Proposing a scalable zero-shot framework that converts static 3D scenes into simulation-ready interactive replicas with consistent geometry, leveraging open-vocabulary object detection and articulation estimation to identify potential interactive parts and recover their motion dynamics.

Contributions and Results:

- Conceptualized and implemented the entire pipeline.
- Achieved state-of-the-art performance with a large margin over strong baseline methods.

Impact: Provides a novel solution for generating interactive scenes directly from existing static datasets, substantially reducing labor-intensive annotations and advancing research in robotics and embodied intelligence.

Research Areas: Scene Understanding, 3D Vision, Instance and Part Segmentation, Articulation Estimation, Robotics.

Deep Reinforcement Learning for Microrobot Navigation in vivo, Master Thesis in ETH ARSL 07/2025 – present

Supervisor: *Prof. Dr. Daniel Ahmed, Dr. Mahmoud Medany* **Role:** *Main Contributor*

Project Introduction: Researching and developing a deep reinforcement learning framework for ultrasound-driven microrobots, training agents in 2D and 3D vascular simulations with the goal of transferring to in vivo experiments, aiming to achieve closed-loop autonomous navigation in mouse brain vasculature for applications such as targeted drug delivery and clot removal.

Contributions and Results:

- Implemented a hybrid model using A* for path planning and PPO for control, significantly accelerating convergence.
- Built 3D vascular maps from 2D angiographic images and designed a 3D training environment for agent RL.
- To be submitted to ***Science Robotics*** after completion of experiments in mouse brain vasculature.

Research Areas: Deep Reinforcement Learning, Microrobotics, AI for Science.

Understanding Collaborative Human Activity from Multiple Minds and Views, ETH CVG **09/2024 – present**
Supervisor: *Prof. Dr. Marc Pollefeys*, *Prof. Dr. Xi Wang* **Role:** *Co-first Author*
Project Introduction: Creating a novel egocentric dataset capturing unscripted multi-agent collaboration in cooking scenarios. It integrates multimodal sensory data with detailed collaboration annotations, enabling research on collaborative behaviors, interaction dynamics, and downstream tasks in human teamwork, robotics, and AI.
Contributions and Results:
- Designed and implemented multimodal data collection, scalable annotation, and cross-view synchronization pipelines.
- Defined and categorized collaboration cues to guide systematic annotation and behavior analysis.

Research Areas: Egocentric Perception, Multi-Agent Collaboration, AI Assistants, Computer Vision, Dataset Design.

Automating Visual Inventory of Building Products, ETH AI Center **02/2025 – 06/2025**
Supervisor: *Prof. Dr. Pei-Yu Wu*, *Prof. Dr. Mennatallah El-Assady* **Role:** *Co-first Author*
Project Introduction: Building a hybrid vision–language pipeline for automated on-site building product inventories, combining object detection with VLM reasoning and expert-in-the-loop refinement across egocentric/indoor videos.
Contributions and Results:
- Designed and implemented the pipeline with an interactive interface for automated visual building inventories.
- Assembled a multi-modal dataset and established a structured annotation process.
- Achieved strong performance across evaluations, demonstrating the feasibility of AI-assisted building audits.

Research Areas: Computer Vision, VLM, Interactive ML, Automation in Construction, Dataset Design.

AI 4 Everyone (Vertically Integrated Projects), UNSW **02/2022 – 12/2022**
Supervisor: *Prof. Dr. Maurice Pagnucco*, *Prof. Dr. Yang Song* **Role:** *Member of RoboCup@Rescue*
Project Introduction: Developing a robotic agent for the RoboCup Rescue Virtual Robot Simulation League, focusing on navigation, perception, and victim detection in disaster scenarios.
Key Technologies: Robotics, Computer Vision, Deep Learning

INTERNSHIP EXPERIENCE

Tencent Technology (Shenzhen) Co., Ltd., Tencent Quantum Laboratory **09/2021 – 07/2022**
Supervisor: *Prof. Dr. Shengyu Zhang*, *Xiong Xu*, *Dr. Yu Zhou* **Position:** *R&D Intern*
Project Introduction (04/2022 - 07/2022): Researching and developing EDA auto-routing software for superconducting quantum chips, including automated air bridge placement to suppress parasitic modes, layout optimization for dense structures, and etch compensation algorithms to enhance quantum gate fidelity and circuit reliability.
Contributions and Results:
- Conducted research and implemented topological algorithms for automatic routing in quantum chip design.
- Authored and published patents based on novel algorithms.

Research Areas: EDA for Quantum Chips, Automated Routing Algorithms, Topological and Geometric Methods.

Project Introduction (01/2022-04/2022): Researching and developing a quantum compiler and programming language for quantum computers.
Responsibilities and Contributions:
- Conducted research and analysis on compiler principles, design, and LLVM-based architecture.
- Designed the CFG grammar and instruction set architecture (ISA) of a quantum compiler using MLIR.

Research Areas: Compiler Design, LLVM, Programming Languages, Quantum Compilation.

Project Introduction (09/2021-12/2021): Developing an ELK-based monitoring and alert system for quantum computers and dilution refrigerators, providing real-time data collection, cleaning, integration, and visualization through an internal web system.
Responsibilities and Contributions:
- Designed, developed, deployed, and maintained the whole system, including monitoring, automated alerts, and web-based visualization.

Research Areas: System Design, Data Integration and Visualization, ELK Stack, Cloud Deployment

TECHNICAL SKILLS

- Proficiency in Python, Java, C/C++, Go, PyTorch, Linux, Git, JavaScript, HTML/CSS, SQL, etc.
- Language: English (fluent), Chinese (native)